CSC 421 - Uncertainty 2

Probabilistic Reasoning

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Readings

Chapter 13

1. Basic: Sections **13.1, 13.2**, **13.3**, **13.4**, and **Summary**
2. Expected: same as Basic
3. Advanced: All the chapter including bibliographical and historical notes

Be sure you are familiar with the pseudocode. You may be tested on your understanding of it in the final.

Last Class

- Notation of Probability
- Joint Probability: $P(X_1, X_2, \ldots, X_n) \equiv P(X_1 \wedge X_2 \wedge \cdots \wedge X_n)$
- Indenpendence
  - a, b independent, then $P(a, b) = P(a)P(b)$
- Probability Rules

Rule	for two events a, b
Negation	$P(\neg a) = 1 - P(a)$
Inclusion-Exclusion	$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$
Marginalization	$P(a) = P(a, b) + P(a, \neg b)$
Conditioning	$P(a) = P(a\|b)P(b) + P(a\|\neg b)P(\neg b)$

- Conditional Probability: the probability of a given b.

$$P(a|b) = \frac{P(a \wedge b)}{P(b)}$$

- Baye's Rule

$$P(a|b) = \frac{P(b|a)P(a)}{P(b)}$$

- Naïve Bayes Model:

$$P(C, E_1, \ldots, E_n) = P(C) \prod_i P(E_i|C)$$

  - Assumes all the E_i independent of each other given C
  - It is a special case of Bayesian Networks (we will introduce today)
- Bayes Classification

$$\hat{y} = \arg\max_y \; P(y|X) = \arg\max_y \; P(X|y)P(y)$$

  - derived from Baye's Rule
  - Naïve Bayes Model can be used to calculate $P(X|y)$ when satisfy its assumption
    - Y can be viewed as the set of all candidate causes (to be classified), X as all the effects.
    - find the "best $Cause$" given all $Effect$

$$P(X|y) = \frac{P(X, y)}{P(y)} = \frac{P(X_1, X_2, \ldots, X_n, y)}{P(y)} = \frac{P(y) \prod_i P(X_i|y)}{P(y)}$$

🤔 Consider: How to represent and calculate joint probability distribution like this

$$P(X_1, X_2, \ldots, X_n),$$

where any X_i, X_j may be dependent of each other.

❗ Naive Bayes model no longer works!

If we assume every X_i is dependent with all r.v. in $\{X_1, \ldots, X_{i-1}\}$ then by chain rule

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i|X_1, \ldots, X_{i-1})$$

But in reality, we usually have:

$$X_i \perp some\ variables \mid some\ others$$

4. Bayesian Networks

- A Bayesian network (BN) is a graphical notation that represents the dependencies among random variables.
  - BN is a Probabilistic Graphical Model (PGM)
- Bayesian networks have the following properties:
  - Directed acyclic graphs (DAGs).
  - Each node = a random variable (RV).
  - $Z \rightarrow X$ in the graph means Z is a parent of X. That is, prob distribution of X depends on Z.
  - Each node X has conditional probability distribution $P(X|Parents(X))$.

Conditional Probability Table (CPT)

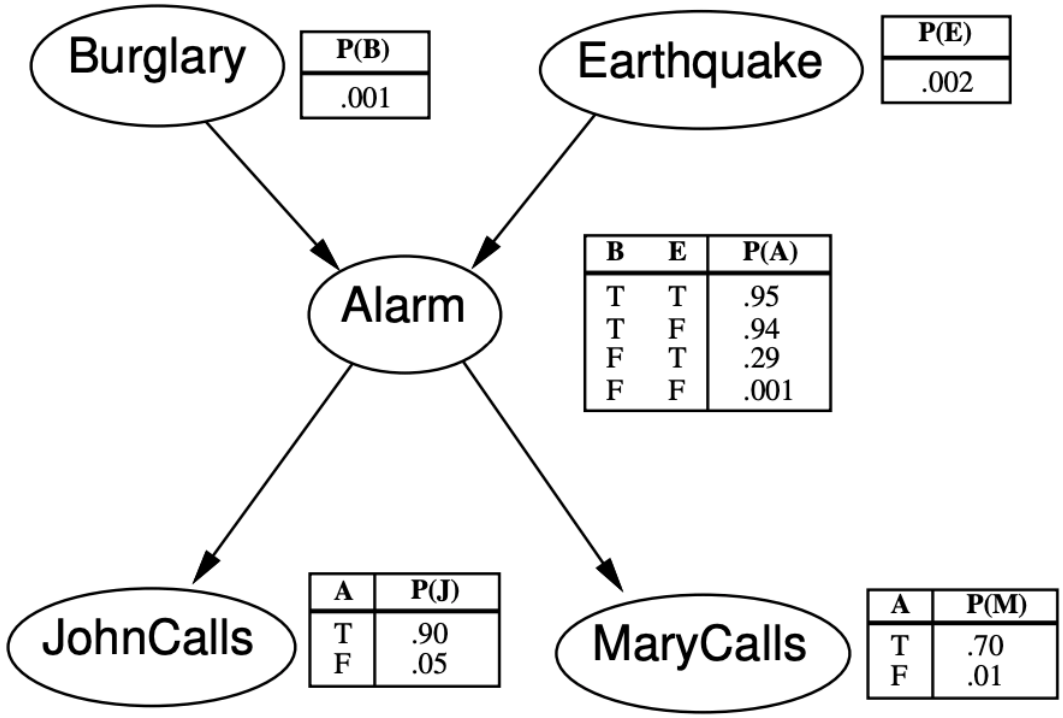
In the simplest case, conditional distribution is represented as a conditional probability table (CPT) giving the distribution over X_i for each combination of parent values.

- A CPT for Boolean X_i with k Boolean parents has 2^k rows for the combinations of parent values

Example: burglary network

I'm at work, neighbor John calls to say my alarm is ringing, but neighbor Mary doesn't call. Sometimes it's set off by minor earthquakes. Is there a burglar?

- Variables: $Burglar$, $Earthquake$, $Alarm$, $JohnCalls$, $MaryCalls$
- Network topology reflects "causal" knowledge
  - A burglar can set the alarm off
  - An earthquake can set the alarm off
  - The alarm can cause Mary to call
  - The alarm can cause John to call



Semantics

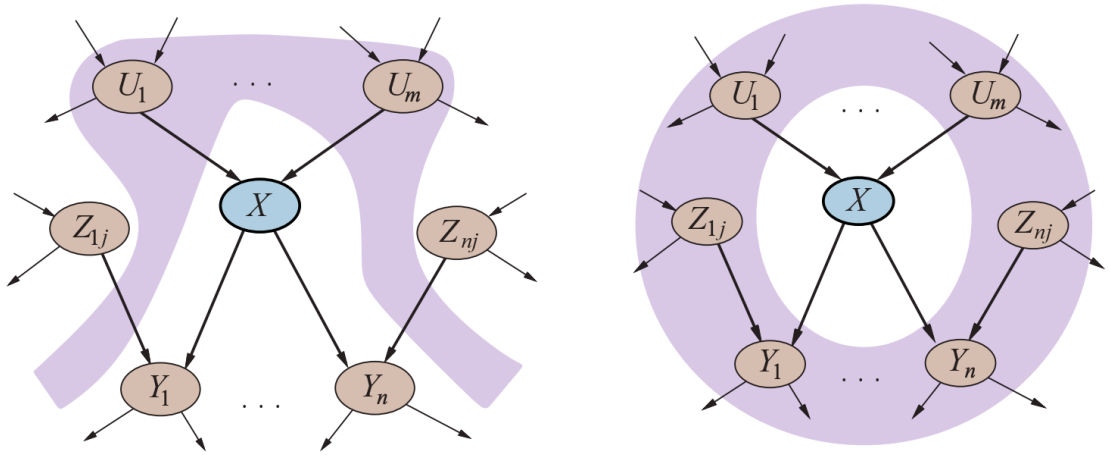
$$\begin{aligned}\text{e.g., } P(j \wedge m \wedge a \wedge \neg b \wedge \neg e) &= P(j \mid a)\, P(m \mid a)\, P(a \mid \neg b, \neg e)\, P(\neg b) \\ &= 0.9 \times 0.7 \times 0.001 \times 0.999 \times 0.998 \\ &\approx 0.00063\end{aligned}$$

Global semantics defines the full joint distribution as the product of the local conditional distributions.

- $$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid \text{parents}(X_i))$$

❓ Q: For a BN $A \to B \to C$, is there any dependency between A and C? How do we model it?

- **Local semantics:** each node is conditionally independent of its nondescendants ($Z_{i,j}$) given its parents (U_i in the gray area)
- **Markov blanket** = parents + children + children's parents.
  - Each node is conditionally independent of all others given its Markov blanket.



❓ Q: **For a BN $A \rightarrow B \leftarrow C$, is there any dependency between A and C? Why?**

❓ Q: **Can we say "Each node is conditionally independent of all others given its parents and children"? Why?**

Constructing Bayesian networks

- **Algorithm:** a series of locally testable assertions of conditional independence guarantees the required global semantic

1. Choose an ordering of variables $X_1, \ldots, X_n$
2. For $i = 1$ to n
   add X_i to the network
   select parents from $X_1, \ldots, X_{i-1}$ such that
   $$\mathbf{P}(X_i | Parents(X_i)) = \mathbf{P}(X_i | X_1, \ldots, X_{i-1})$$

(see demo)

5. Probabilistic reasoning in Bayesian networks

In probabilistic reasoning, we want to know $P(y|X)$ for each y. Ideally, we can use BN to calculate $P(X, y)$.

$$P(y|X) = \frac{P(X, y)}{P(X)} = \frac{P(X_1, X_2, \ldots, X_n, y)}{P(X)}$$

In reality, we face the challenge of **hidden variables**.

- BN typically contains variables that are not observed in a given problem.

They are called hidden variables Z.

- What we need for classification is $P(X, y)$. However, the Bayesian network defines only the full joint distribution $P(y, X, Z) = \prod_V P(V \mid Parents(V))$.
- In this case, we must marginalize (sum out) those hidden variables.

$$P(y, X) = \sum_Z P(y, X, Z)$$

- This summation over hidden variables is what makes inference non-trivial.

5.1 Exact Inference

Let's use formal notation, goal is calculate $P(Query \mid Evidence)$, i.e., $P(X|e)$. Y denotes hidden variables.

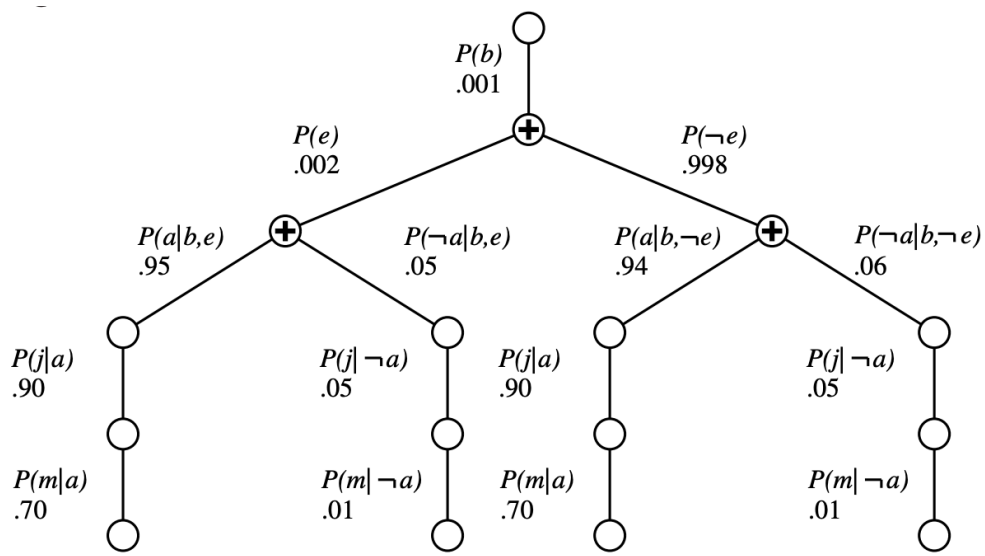
- **Enumeration**

$$P(X|e) = \alpha P(X, e) = \alpha \sum_y^Y P(X, e, y)$$

  - See text book for the pseudocode
  - Recursive depth-first enumeration: $O(n)$ space, $O(d^n)$ time
  - inefficient: repeated computation
  - Example: burglary network

$$\begin{aligned} P(B|j,m) &= \frac{P(B,j,m)}{P(j,m)} \\ &= \alpha P(B,j,m) \\ &= \alpha \textstyle\sum_e \sum_a P(B,j,m,a,e) \end{aligned}$$

$$\begin{aligned} P(B|j,m) &= \alpha \textstyle\sum_e \sum_a P(B)P(e)P(a|B,e)P(j|a)P(m|a) \\ &= \alpha P(B) \textstyle\sum_e P(e) \sum_a P(a|B,e)P(j|a)P(m|a) \end{aligned}$$



- **Variable elimination**
  - See text book for pseudocode
  - Carry out summations right-to-left, storing intermediate results (factors) to avoid recomputation
  - **Factor**: Each CPT is a factor $f(X, Parents(X))$
  - **Pointwise product** $\times$: $f(X,Y) \times g(Y,Z) = h(X,Y,Z)$
  - Example: burglary network

$$
\begin{aligned}
P(B|j,m) &= \alpha \underbrace{P(B)}_{B} \sum_e \underbrace{P(e)}_{E} \sum_a \underbrace{P(a|B,e)}_{A} \underbrace{P(j|a)}_{J} \underbrace{P(m|a)}_{M} \\
&= \alpha f_B(B) \times \sum_e f_E(E) \times \sum_a f_A(A,B,E) \times f_J(A) \times f_l \\
&= \alpha f_B(B) \times \sum_e f_E(E) \times f_{\overline{A}JM}(B,E) \quad \text{(sum out } A\text{)} \\
&= \alpha f_B(B) \times f_{\overline{EA}JM}(B) \quad \text{(sum out } E\text{)}
\end{aligned}
$$

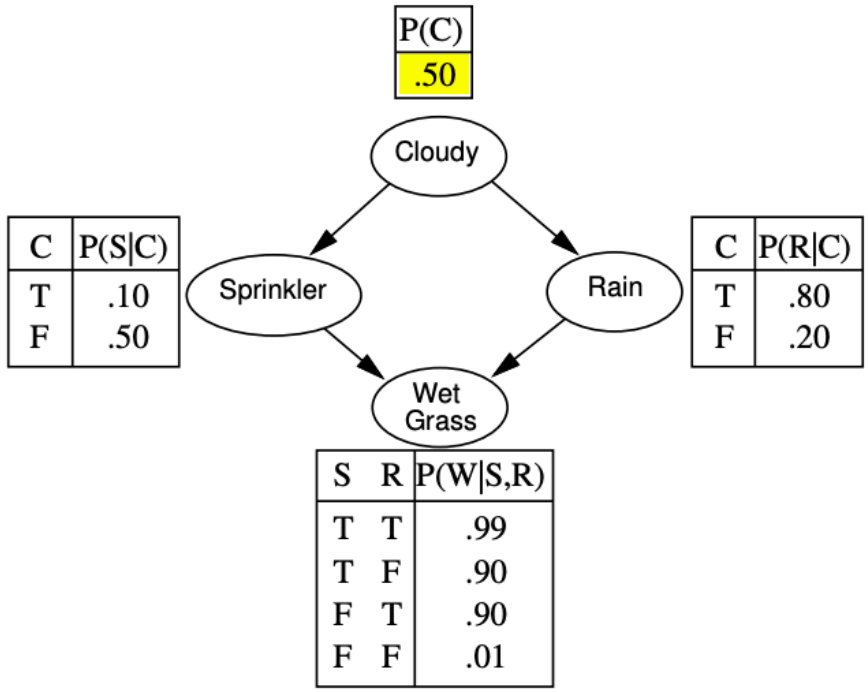
5.2 Approximate Inference

Since exact inference is NP-hard, we consider randomized sampling, a.k.a. Monte Carlo algorithms. See textbook for pseudocode and formal formulas.

- **Direct Sampling**:
  - Prior sampling: Sample variables in turn, in topological order, until all variables have values. Repeat N times. Estimate the joint prob distribution with:

$$P(X = x) \approx \frac{\#\text{ of }(X = x)}{N}$$

- see textbook for the pseudocode.



- Rejection sampling: reject samples disagreeing with evidence

$$P(X = x \mid e) \approx \frac{\#\text{ of }(X = x \wedge e)}{\#\text{ of }(e)} = \hat{P}(X = x \mid e)$$

  - see textbook for the pseudocode.
  - Estimate $P(Rain|Sprinkler = true)$ using 100 samples
    - 27 samples have $Sprinkler = true$. Of these, 8 have $Rain = true$ and 19 have $Rain = false$.

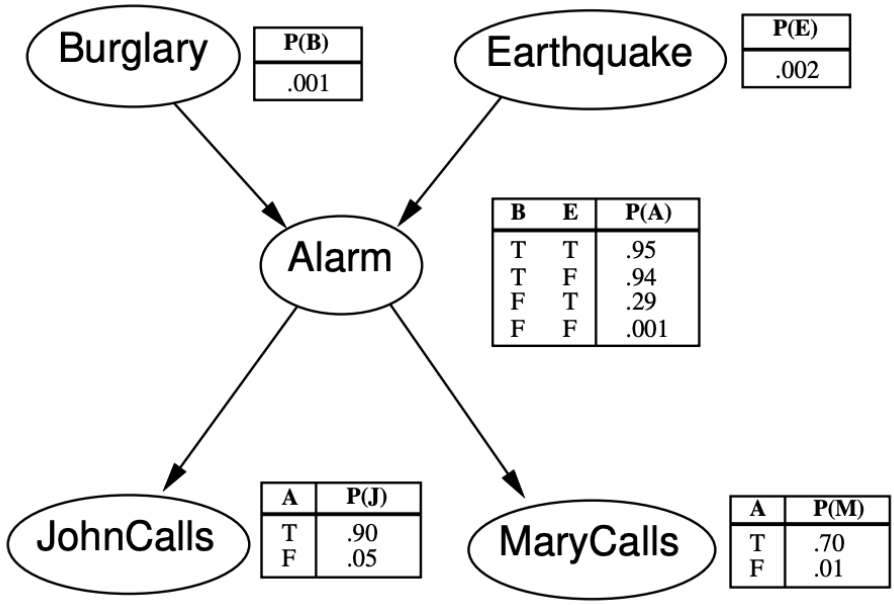
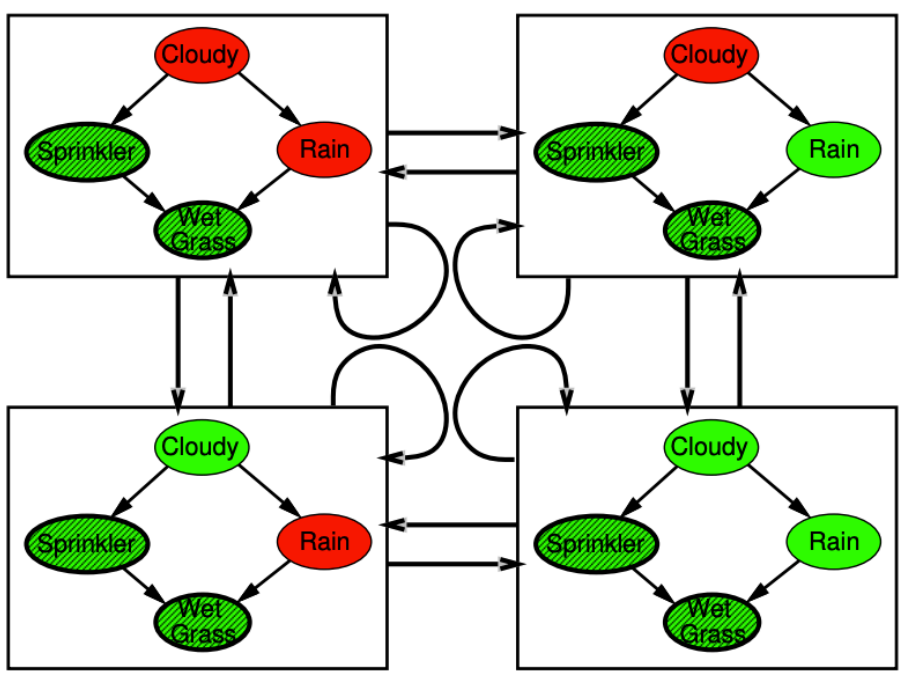
      $$\hat{P}(Rain|Sprinkler = true) = Normalize(\langle 8, 19 \rangle) = \langle 0.296, 0$$

    - Similar to a basic real-world empirical estimation procedure
    - Problem: if $P(e)$ is small, most samples will be rejected, making the method very inefficient.
- Likelihood weighting: use CPT of evidence to weight samples, instead of rejecting samples

$$P(X = x \mid e) \approx \frac{\sum_i^N w_i \mathbf{1}(X_i = x)}{\sum_i^N w_i},$$

$$\text{where } w_i = \prod_{E_j \in evidence} P(e_j \mid parents(E_j))$$

- see textbook for the pseudocode.



- **Markov chain Sampling** (MCMC)
  - Instead of generating each sample from scratch, MCMC algorithms generate each sample by making a random change to the preceding sample.
  - Gibbs sampling: sampling variables given their Markov blankets.
    - see textbook for the pseudocode.